Project: Worked examples
Title: Pipe expansion, loops and anchor forces (screening)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Screening only — not a piping stress analysis. Thermal growth of a straight run between anchors, the fully restrained stress and force, the leg an offset leg or a U-loop (two guided cantilevers) needs to absorb the growth within the owner's allowable displacement-stress range Sa and whether the available leg is long enough (an offset leg or Z-bend also loads its run or runs with the leg's own growth), the force on the anchors and the largest root moment of the guided-cantilever model, the screening ratio $D \cdot Y/(L - U)^2$ against the owner's K1 (PASS, or the line needs a formal stress analysis), and, for an axial bellows, the anchor force P·A_eff plus its spring force. No allowable stress, K1 or code table ships.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Pipe section | $d_i = OD - 2 \cdot t$; $A_m = \pi/4 \cdot (OD^2 - d_i^2)$; $I = \pi/64 \cdot (OD^4 - d_i^4)$; $Z = I/(OD/2)$ |
| Temperature change | $\Delta T = T_{operate} - T_{install}$, or as entered; the movement uses $\lvert\Delta T\rvert$ and its sign gives the direction |
| Run that grows | $L_{tot} = L$, or $L + L2$ for a Z-bend; thermal growth $\Delta L = \alpha \cdot L_{tot} \cdot \lvert\Delta T\rvert$ |
| Fully restrained pipe | $\sigma_r = E \cdot \alpha \cdot \lvert\Delta T\rvert$ (E at the installation temperature); $F_r = E_r \cdot \alpha \cdot \lvert\Delta T\rvert \cdot A_m$ ($E_r = E_{hot}$, else E) |
| Movement one leg must absorb | $\Delta = \Delta L$ for an offset leg or Z-bend; $\Delta = \Delta L/2$ for a U-loop (two identical legs in series, the cross leg a rigid guide) |
| Guided cantilever, required leg | $L_{req} = \sqrt{(3 \cdot E \cdot OD \cdot \Delta/Sa)}$, the leg at which the bending stress M/Z reaches Sa; PASS when $L_{req} \le$ the available leg H |
| Force and moment of the available leg | $F_{leg} = 12 \cdot E_r \cdot I \cdot \Delta/H^3$ (acts along the run); $M_{leg} = 6 \cdot E_r \cdot I \cdot \Delta/H^2$ ($= F_{leg} \cdot H/2$); $\sigma_{leg} = 3 \cdot E \cdot OD \cdot \Delta/H^2 = Sa \cdot (L_{req}/H)^2$ |
| Offset leg or Z-bend, the runs bend for the leg's growth $\Delta_{leg} = \alpha \cdot \lvert\Delta T\rvert \cdot H$ | the runs act in series under one force $F_{run} = 12 \cdot E_r \cdot I \cdot \Delta_{leg}/(L^3 + L2^3)$ ($L2 = 0$ for an L-bend) acting along the leg; run i takes the share $L_i^3/(L^3 + L2^3)$ of $\Delta_{leg}$, with root stress $\sigma_i = 3 \cdot E \cdot OD \cdot \Delta_i/L_i^2$ and moment $M_i = F_{run} \cdot L_i/2$; each run is checked as its own cantilever, PASS when $\sqrt{(\sigma_i/Sa)} \le 1$ (for an L-bend the run needed is $L_{req,run} = \sqrt{(3 \cdot E \cdot OD \cdot \Delta_{leg}/Sa)}$) |
| Force on the anchors | the two components are perpendicular, $F_{leg}$ along the run and $F_{run}$ along the leg, so the force is $\sqrt{(F_{leg}^2 + F_{run}^2)}$; the moment reported is the largest root moment of the legs and runs in the guided-cantilever model |
| Flexibility screening | $OD \cdot Y/(L_{dev} - U)^2 \le K1$ with $Y = \alpha \cdot \lvert\Delta T\rvert \cdot U$, the free growth between the anchors; offset leg or Z-bend $L_{dev} = L_{tot} + H$, $U = \sqrt{(L_{tot}^2 + H^2)}$; U-loop $L_{dev} = L + 2 \cdot H$, $U = L$ (so $Y = \Delta L$); PASS, or the line needs a formal stress analysis |

| WHAT IT CALCULATES | RELATION |
|---|---|
| Axial bellows (no leg or loop, no leg check, no screening) | anchor force = $P \cdot A_{eff} + k \cdot \Delta L$; the movement is checked against the bellows' rated axial movement when given |

Basis: Process piping — flexibility analysis, exemption from formal analysis[1](para. 319.4.1 (screening relation; number to be checked against the owner's copy)); Process piping — displacement stress range and reactions[1](para. 319.3.2 and 319.5 (reference and hot moduli; numbers to be checked against the owner's copy)); Metallic industrial piping — flexibility analysis[2](cl. 12 (number to be checked against the owner's copy)).

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| E | Young's modulus of the pipe material at the installation (reference) temperature, for the stress check, in GPa (steel 210, copper 120, PE and PP about 1; not MPa) | 210 | GPa | assumption: W3 seed 'Carbon steel (generic)', indicative; replace with the project's material record. | 0.5 ... 500 GPa |
| α | Mean coefficient of thermal expansion between the install and operating temperatures | 12 | um/(m·K) | assumption: W3 seed 'Carbon steel (generic)' (12e-6 1/K), indicative; replace with the project's material record. | 1 ... 250 um/(m·K) |
| OD | Pipe outside diameter | 88.9 | mm | drawing: DRG-M-210 rev B | 10 ... 1500 mm |
| $t_{wall}$ | Pipe wall thickness | 3.2 | mm | drawing: DRG-M-210 rev B | 0.5 ... 60 mm |
| L | Anchor spacing: the straight length that grows (an offset leg or Z-bend: the first run, which also bends for the leg's growth) | 50 | m | drawing: DRG-M-210 rev B | 1 ... 500 m |
| $temp_{mode}$ | Temperature change from | install and operating temperatures | | assumption: Install and operating temperatures are both known. | any |
| $T_{install}$ | Temperature at installation (when the run is anchored) | 10 | °C | assumption: Mean site temperature when the main is anchored. | -50 ... 50 °C |
| $T_{operate}$ | Operating temperature (hot or cold) | 80 | °C | spec: SPEC-M-04 §3: LTHW flow temperature | -50 ... 500 °C |
| layout | What absorbs the movement | U-loop (two legs, each takes half the movement) | | drawing: DRG-M-210 rev B | any |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| H | Available leg length (each leg of a U-loop; an offset leg: the leg off the run, which may be the longer of the two); no default, not needed for an axial bellows | 4 | m | drawing: DRG-M-210 rev B | 0.5 … 50 m |
| Sa | Allowable displacement-stress range Sa for the legs and runs (your piping code and its formula; no default; not needed for an axial bellows) | 100 | MPa | assumption: Invented illustrative allowable range; take Sa from the project's piping code and its formula. | 5 … 500 MPa |
| $K_{scr}$ | Screening limit K1 for OD·Y/(L − U)², in mm²/m² (or in²/ft², …) from your piping code (no default; blank = screening not evaluated and the sheet does not pass) | 120 | mm²/m² | assumption: Invented illustrative screening limit; take K1 from the project's piping code. | 1 … 1000 mm²/m² |

### 3. Assumptions

A1. Screening only — not a piping stress analysis. The result tells whether a simple leg or loop is plausible and what it puts on its anchors; it does not replace a flexibility analysis of the real routing (elbow flexibility, intensification, torsion, branches, restraint friction, supports, sustained and occasional loads). (Stated scope.)

A2. A guided cantilever: the available leg is straight, uniform, fixed against rotation at both ends and loaded only by the movement it absorbs (stiffness $12 \cdot E \cdot I / H^3$, end moment $F \cdot H/2$). Elbows are taken as rigid and the leg as slender: the beam model is not valid for legs shorter than about ten diameters. (Beam theory.)

A3. No stress-intensification factor and no elbow flexibility are applied: the stress at the elbow root of a real bend is higher than the beam stress M/Z, by the intensification factor of the code, so a leg that passes here can still exceed Sa at the elbow. This is the main reason the result is a screen and not a stress analysis. (Stated limit.)

A4. A U-loop is two identical legs in series, each a guided cantilever taking half the growth, the cross leg a rigid guide; the force along the line is the same F in both legs. The moment M is the moment at the leg roots (the corners of the loop); it reaches an anchor only where the leg ends at the anchor (an offset leg). The growth of the loop's own legs, which the cross leg has to absorb, is not checked. (Loop model.)

A5. An offset leg is anchored at both ends: the run L bends for the growth of the leg ($\alpha \cdot |\Delta T| \cdot H$) as a second guided cantilever and is checked against the same Sa. A Z-bend (second run L2) has the two runs in series under one force, each taking its share of the leg's growth in proportion to its flexibility, $L_i^3$, and each checked as its own cantilever; the leg takes the growth of both runs. The force on the anchors is the resultant of the force along the run (the leg's) and the force along the leg (the runs'), which are perpendicular. (Offset-leg and Z-bend model.)

A6. $anchor_{moment}$ is the largest root moment of the legs and runs in the guided-cantilever simplification ($M = F \cdot L/2$ with each end fixed against rotation). It is not the moment on a frame anchor, which depends on the whole frame (the stiffness of the other legs, the elbow rotations, the anchor's own fixity); a formal analysis is needed before an anchor is sized for moment. (Stated limit.)

A7. The material properties are single values: α the mean coefficient between the install and operating temperatures; E the modulus at the installation (reference) temperature, used for the stress check, as the piping codes do (ASME B31.3 para. 319.3.2 and 319.5, numbers to be checked against the owner's copy); $E_{hot}$, when given, is used only for the reactions on the anchors. Take them from a W3 material applied at

the right temperature. (Material library; code convention to be checked.)
A8. Sa and K1 are the owner's records from the owner's own copy of the piping code (or the design basis), with their provenance; no allowable stress, K1 or code constant ships. The sheet applies the relation to the numbers entered and does not know which code, material or load case they belong to. (No table from a standard is shipped.)
A9. No friction, weight, spring-hanger or cold-spring effect is included in the leg force; the anchor force adds the friction of the sliding supports, μ·w·L, only when μ and the weight per metre are entered (a warning says so when they are not). The axial bellows anchor force is the pressure thrust $P \cdot A_{eff}$ plus the spring force k·ΔL at the governing pressure entered (the design or the test pressure, whichever governs; a bellows' compression and extension ratings can differ): guide friction and any other bellows load come from the manufacturer's data. (Stated limit.)
A10. E (Young's modulus of the pipe material at the installation (reference) temperature, for the stress check, in GPa (steel 210, copper 120, PE and PP about 1; not MPa)) is taken as 210 GPa. (W3 seed 'Carbon steel (generic)', indicative; replace with the project's material record.)
A11. α (Mean coefficient of thermal expansion between the install and operating temperatures) is taken as 12 um/(m*K). (W3 seed 'Carbon steel (generic)' (12e-6 1/K), indicative; replace with the project's material record.)
A12. $temp_{mode}$ (Temperature change from) is taken as install and operating temperatures. (Install and operating temperatures are both known.)
A13. $T_{install}$ (Temperature at installation (when the run is anchored)) is taken as 10 degC. (Mean site temperature when the main is anchored.)
A14. Sa (Allowable displacement-stress range Sa for the legs and runs (your piping code and its formula; no default; not needed for an axial bellows)) is taken as 100 MPa. (Invented illustrative allowable range; take Sa from the project's piping code and its formula.)
A15. $K_{scr}$ (Screening limit K1 for OD·Y/(L − U)², in mm²/m² (or in²/ft², ...) from your piping code (no default; blank = screening not evaluated and the sheet does not pass)) is taken as 120 mm^2/m^2. (Invented illustrative screening limit; take K1 from the project's piping code.)
A16. Layout: the U-loop is two legs of 4 m in series, each a guided cantilever taking half the movement (21 mm); the cross leg is a rigid guide and its width does not enter. Screening only — not a piping stress analysis. (Guided-cantilever model.)

## 4. Working

[9] $dT = T_{operate} - T_{install}$
$= 80\ degC - 10\ degC$
$= 70\ K$
Temperature change, operating − install
[11] $E_r = E$
$= 210\ GPa$
$= 210\ GPa$
Modulus for the reactions (E, no $E_{hot}$ given)
[12] $D_i = OD - 2 * t_{wall}$
$= 88.9\ mm - 2 \times 3.2\ mm$
$= 82.5\ mm$
Inside diameter
[13] $A_m = pi / 4 * (OD^2 - D_i^2)$
$= \pi / 4 \times ((88.9\ mm)^2 - (82.5\ mm)^2)$
$= 861.55\ mm^2$
Metal cross-section
[14] $I_x = pi / 64 * (OD^4 - D_i^4)$
$= \pi / 64 \times ((88.9\ mm)^4 - (82.5\ mm)^4)$
$= 79.206\ cm^4$
Second moment of area of the pipe section

[15] $Z_x = I_x / (OD / 2)$
= 79.206 cm⁴ / (88.9 mm / 2)
= 17.819 cm³
Section modulus
[16] $L_{tot} = L$
= 50 m
= 50 m
Length that grows between the anchors
[17] $dL = \alpha * L_{tot} * abs(dT)$
= 12 um/(m·K) × 50 m × abs(70 K)
= 42 mm
Thermal growth between the anchors, as a magnitude
[18] $dL_{signed} = \alpha * L_{tot} * dT$
= 12 um/(m·K) × 50 m × 70 K
= 42 mm
Thermal movement with its sign: positive lengthens, negative shortens
[19] $\sigma_{res} = E * \alpha * abs(dT)$
= 210 GPa × 12 um/(m·K) × abs(70 K)
= 176.4 MPa
Stress in the fully restrained pipe (E at the installation temperature)
[20] $F_{res} = E_r * \alpha * abs(dT) * A_m$
= 210 GPa × 12 um/(m·K) × abs(70 K) × 861.55 mm²
= 151.98 kN
Force on the anchors of the fully restrained pipe
[21] $F_{friction}$ = support friction not entered
= μ and $w_{pipe}$ blank
= 0 N
Friction of the sliding supports (not entered: none in the anchor force)
[24] $u_{res} = \sigma_{res} / Sa$
= 176.4 MPa / 100 MPa
= 1.764
Fully restrained stress as a fraction of Sa (above 1: the movement has to be absorbed)
[25] $\Delta = dL / 2$
= 42 mm / 2
= 21 mm
Movement each leg of the U-loop absorbs (half of it)
[26] $L_{req} = sqrt(3 * E * OD * \Delta / Sa)$
= sqrt(3 × 210 GPa × 88.9 mm × 21 mm / 100 MPa)
= 3.4295 m
Leg length at which the guided-cantilever bending stress reaches Sa
[27] $u_{leg} = L_{req} / H$
= 3.4295 m / 4 m
= 0.85737
Required leg over available leg
[28] $F_{leg} = 12 * E_r * I_x * \Delta / H^3$
= 12 × 210 GPa × 79.206 cm⁴ × 21 mm / (4 m)^3
= 654.93 N
Force of the available leg on its anchors or guides
[29] $M_{leg} = 6 * E_r * I_x * \Delta / H^2$
= 6 × 210 GPa × 79.206 cm⁴ × 21 mm / (4 m)^2
= 1309.9 N·m
Moment at the root of the available leg

[30] $\sigma_{leg}$ = 3 * E * OD * Δ / H^2
= 3 × 210 GPa × 88.9 mm × 21 mm / (4 m)^2
= 73.509 MPa
Bending stress at the root of the available leg (E at the installation temperature)
[31] $u_{sigma}$ = $\sigma_{leg}$ / Sa
= 73.509 MPa / 100 MPa
= 0.73509
Bending stress over Sa (= $u_{leg}^2$)
[32] $F_{along,run}$ = $F_{leg}$ + $F_{friction}$
= 654.93 N + 0 N
= 654.93 N
Component of the anchor force along the run: the loop's force plus the support friction
[33] $F_{along,leg}$ = no component along the leg
= a U-loop is in line
= 0 N
Component of the anchor force along the leg: none for a U-loop
[34] $anchor_{force}$ = $F_{along,run}$
= 654.93 N
= 654.93 N
Force on the anchors (the force along the loop; support friction only if μ and $w_{pipe}$ are entered)
[35] $anchor_{moment}$ = $M_{leg}$
= 1309.9 N·m
= 1309.9 N·m
Moment at the root of the loop's legs
[36] $L_{dev}$ = L + 2 * H
= 50 m + 2 × 4 m
= 58 m
Developed length between the anchors, run plus two legs
[37] $U_a$ = L
= 50 m
= 50 m
Straight distance between the anchors (the loop is in line)
[38] $Y_{scr}$ = α * abs(dT) * $U_a$
= 12 um/(m·K) × abs(70 K) × 50 m
= 42 mm
Resultant displacement Y: the free growth between the anchors
[39] scr = OD * $Y_{scr}$ / ($L_{dev}$ - $U_a$)^2
= 88.9 mm × 42 mm / (58 m − 50 m)^2
= 58.341 mm²/m²
Screening ratio OD·Y/(L − U)²
[41] $u_{scr}$ = scr / $K_{scr}$
= 58.341 mm²/m² / 120 mm²/m²
= 0.48617
Screening ratio over the K1 record
[42] pass = K1 given and every check passes
= 2 check(s)
= 1
1 when the leg, the run (offset leg) and the screening checks all pass; 0 when one fails or when no K1 was entered

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $D_i$ | 82.5 mm | Inside diameter |
| $A_m$ | 861.55 mm² | Metal cross-section |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $I_x$ | 79.206 cm⁴ | Second moment of area |
| $Z_x$ | 17.819 cm³ | Section modulus |
| dT | 70 K | Temperature change |
| $L_{tot}$ | 50 m | Length that grows |
| dL | 42 mm | Thermal movement between the anchors (magnitude) |
| $dL_{signed}$ | 42 mm | Thermal movement, signed |
| $\sigma_{res}$ | 176.4 MPa | Fully restrained stress |
| $F_{res}$ | 151.98 kN | Fully restrained force |
| direction | "growth: the pipe lengthens; fully restrained it presses on its anchors (compression)" | Direction of the movement |
| $F_{friction}$ | 0 N | Support friction on the run |
| $u_{res}$ | 1.764 | Fully restrained stress over Sa |
| Δ | 21 mm | Movement absorbed by each leg |
| $L_{req}$ | 3.4295 m | Required leg length |
| $u_{leg}$ | 0.85737 | Required leg over available leg |
| $F_{leg}$ | 654.93 N | Force of the leg |
| $M_{leg}$ | 1309.9 N·m | Moment at the leg root |
| $\sigma_{leg}$ | 73.509 MPa | Bending stress at the leg root |
| $u_{sigma}$ | 0.73509 | Bending stress over Sa |
| $F_{along,run}$ | 654.93 N | Anchor force component along the run |
| $F_{along,leg}$ | 0 N | Anchor force component along the leg |
| $anchor_{force}$ | 654.93 N | Force on the anchors (resultant) |
| $anchor_{moment}$ | 1309.9 N·m | Largest root moment (guided-cantilever model) |
| $L_{dev}$ | 58 m | Developed length |
| $U_a$ | 50 m | Anchor distance |
| $Y_{scr}$ | 42 mm | Resultant displacement Y |
| scr | 58.341 mm²/m² | Screening ratio OD·Y/(L − U)² |
| $u_{scr}$ | 0.48617 | Screening ratio over K1 |
| screen | "PASS" | Screening result |
| pass | 1 | All checks pass |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Available leg against the required leg (3.4295 m) | $u_{leg}$ = 0.85737 | $u_{leg} \le 1$ | 0.857 | |
| PASS | Screening only: $OD \cdot Y/(L - U)^2 \le K1$ (not met: the line needs a formal stress analysis) | $u_{scr}$ = 0.48617 | $u_{scr} \le 1$ | 0.486 | [1](para. 319.4.1 (screening form; number to be checked against the owner's copy)) |

## 7. Applicability

WARNING — record: Tabulated values not taken from a reference record: E, α (Material properties); OD, $t_{wall}$ (Pipe outside diameter and wall); Sa (Allowable displacement-stress range); $K_{scr}$ (Screening limit K1). Enter each as one so its source, edition and licence travel with it.
WARNING — friction: μ: the anchor force holds no support friction. Every slide or guide between the anchor and the loop adds μ·w·L to it, which on a long run can exceed the leg force several times; enter μ and the weight per metre (with the length on slides) to include it.

## 8. Verification

```
Template bs.pipe-expansion@1.3.0 Content hash 8dd3b0517fce49e5 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References

1. ASME B31.3.
2. EN 13480-3.